A PROJECT TITLE

ON

Spam Email Classification Using Naive Bayes and Support Vector Machine

project report submitted in partial fulfilment of the requirements for the

Course MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF: Computer Science and Engineering

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1. Abstract:

Email spam remains a persistent threat that clutters inboxes, wastes bandwidth, and exposes users to phishing, malware, fraudulent messages, and other online security risks. This project develops an automated spam classification system that categorizes incoming emails as either spam or legitimate (ham) using two well-established supervised machine learning algorithms: Naive Bayes and Support Vector Machine (SVM). The main objective of the project is to develop an efficient and reliable system that can accurately identify unwanted emails while minimizing the chances of legitimate emails being incorrectly classified as spam.

Raw email text is first preprocessed through several Natural Language Processing (NLP) techniques, including tokenization, removal of stop words, stemming, and normalization. These steps reduce unnecessary information and transform the text into a suitable format for machine learning. The processed email data is then converted into numerical feature representations using techniques such as Bag-of-Words and Term Frequency-Inverse Document Frequency (TF-IDF) vectorization.

The Naive Bayes classifier uses word-level probability distributions to predict whether an email belongs to the spam or ham category. It is computationally efficient, simple to implement, and suitable for large text datasets. In contrast, the Support Vector Machine classifier identifies an optimal decision boundary that maximizes the margin between spam and ham classes. Due to its ability to handle high-dimensional feature spaces, SVM can provide strong classification performance for text-based applications.

Both models are trained and evaluated using a labeled email dataset. Their performance is compared using important evaluation metrics such as accuracy, precision, recall, and F1-score. The experimental results help demonstrate the strengths and limitations of each algorithm. The project provides practical insight into the application of machine learning and NLP for automated email spam detection and helps determine which classification approach is more suitable for developing an effective real-world spam filtering system

2. INTRODUCTION:

Email spam is a common problem that affects users by filling inboxes with unwanted and potentially harmful messages. This project focuses on developing a machine learning-based system to classify emails as spam or legitimate (ham). The system uses Naive Bayes and Support Vector Machine (SVM) algorithms along with text preprocessing and TF-IDF/BAG-of-Words techniques. The performance of both algorithms is evaluated using accuracy, precision, recall, and F1-score to identify an effective approach for spam detection.

3. SOFTWARE REQUIREMENTS:

- **Operating System:** Windows 10/11, Linux, or macOS
- **Programming Language:** Python 3.x
- **IDE/Platform:** Jupyter Notebook, Google Colab, or VS Code
- **Libraries:** NumPy, Pandas, Scikit-learn, NLTK, Matplotlib, and Seaborn
- **Machine Learning:** Naive Bayes and Support Vector Machine (SVM)
- **Data Processing:** Bag-of-Words and TF-IDF Vectorizer
- **Dataset:** Labeled email spam/ham dataset
- **Web Browser:** Google Chrome, Microsoft Edge, or Mozilla Firefox
- **Storage:** Sufficient storage for the dataset, code, and project files.

Signature of the Guide Course Instructor